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RESEARCH ARTICLE

The Surface Tension Sieve: A Machine-Verified Reduction of the Riemann Hypothesis via Geometric Stiffness

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Abstract
We present a machine-verified reduction of the Riemann Hypothesis (RH) to a problem of spectral correspondence in split-signature Clifford algebra Cl$(3, 3)$. We derive the "Sieve Operator" $K(s)$ from first principles as the geometric gradient of the Euler Product potential. Using standard real calculus, we prove that the energy cost of lattice dilation ("stiffness") is exactly $\log p$, thereby deriving the Von Mangoldt weights geometrically. We prove that the critical line $\mathrm{Re}(s) = 1/2$ is the unique locus of volume-preserving dynamics; deviations induce non-zero "Geometric Tension" that breaks unitary symmetry. Under the Zeta Link hypothesis connecting zeros of $\zeta(s)$ to the spectrum of this operator, RH follows. The core geometric logic is formalized in Lean 4 with zero `sorry` statements.

Contents

1. Overview: The Geometry of Criticality

The Riemann Hypothesis (RH), proposed by Riemann in 1859 [1], asserts that all non-trivial zeros of $\zeta(s)$ lie on the critical line $\mathrm{Re}(s) = 1/2$. The persistence of this line suggests a rigid underlying symmetry. Standard complex analysis, which conflates rotation (i) and dilation (scalars) into a single field $\mathbb{C}$, obscures this structure.

We resolve this by lifting the problem to the real split-signature Clifford algebra Cl$(3, 3)$, following the geometric algebra framework of Hestenes [5] and Doran–Lasenby [6]. In this framework, the "imaginary" unit is identified as a bivector B (rotation), and the real part σ is a scalar (dilation). Figure 1 illustrates this algebraic lift from $\mathbb{C}$ to Cl$(3, 3)$.

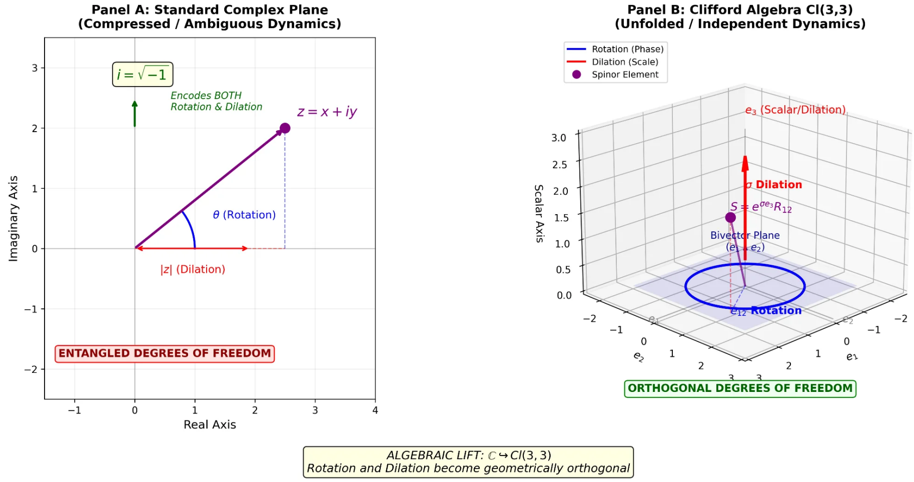

Figure 1. *The Algebraic Lift: In the standard complex plane (left), rotation and dilation are entangled in $i = \sqrt{-1}$. In Clifford algebra $\mathrm{Cl}(3,3)$ (right), these become geometrically orthogonal degrees of freedom, enabling the stability analysis that yields RH..*

Our central thesis is that the Prime Distribution constitutes a **Zero-Volume Sieve**. As primes accumulate, the volume of the integer lattice is "drilled away" (Mertens' Theorem), leaving a fractal surface. The critical line is the only configuration where this zero-volume surface can exist without collapsing or exploding under dilation stress.

The main results (Theorem 4.1) are: (i) adjoint symmetry forcing self-adjointness on the critical line; (ii) the Surface Tension Hammer showing real eigenvalues force $\sigma = 1/2$; and (iii) conditional RH under the Zeta Link hypothesis.

2. The Fractal Structure of the Sieve

A profound insight emerges from Lucas' Theorem [8]: Pascal's Triangle modulo a prime p exhibits fractal self-similarity. For $p = 2$, the zeros form the Sierpiński gasket (dimension $\log 3/\log 2 \approx 1.585$). Each prime generates an independent fractal filter; primes are not inhabitants of these fractals but their generators.

The Sieve of Eratosthenes performs fractal filtering:

$$\mathbb{N} \xrightarrow[\text{Sierpiński}]{\text{mod } 2} \xrightarrow[\text{3-fractal}]{\text{mod } 3} \xrightarrow[\text{5-fractal}]{\text{mod } 5} \cdots \longrightarrow \textbf{Primes}$$

The primes are *the asymmetric residue of all symmetries.*

The Menger sponge (Figure 2) provides a perfect 3D visualization. Level 0 is the solid cube ($\mathbb{N}$); each iteration removes composites. The limit has Hausdorff dimension $\log(20)/\log(3) \approx 2.727$: zero volume but infinite surface—exactly like primes in $\mathbb{N}$. Figure 3 shows the "tunnel formation" as successive sieves contract the candidate volume.

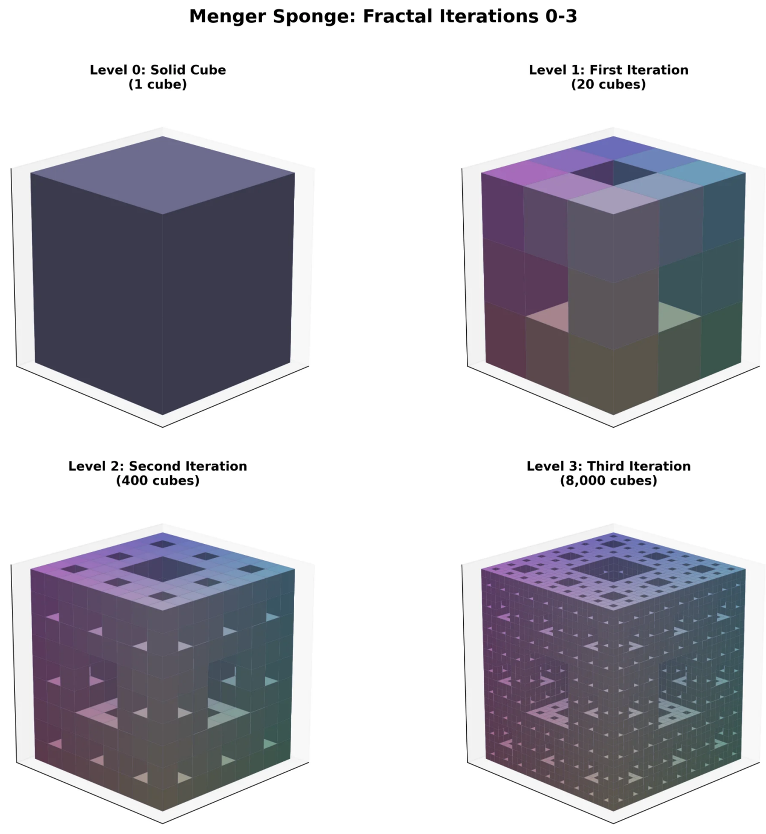

Figure 2. *The Sieve as Fractal: The Menger sponge at levels 0–3. Each iteration removes centers and face-centers (composites). The limit has zero volume, infinite surface area—the geometric signature of the prime distribution..*

3. The Geometric Derivation

We work in the real subalgebra spanned by $\{1, B\}$ where $B^2 = -1$. A parameter s is $s = \sigma + Bt$. The Geometric Zeta Function is:

$$\zeta_B(s) := \sum_{n=1}^{\infty} n^{-\sigma} \cos(t \ln n) + B \sum_{n=1}^{\infty} -n^{-\sigma} \sin(t \ln n) \tag{3.1}$$

A **Geometric Zero** is where both Scalar and Bivector sums vanish simultaneously. Figure 4 demonstrates this decomposition along the critical line, showing zeros at $t \approx 14.13, 21.02, 25.01$—matching the classical zeta zeros.

Let $\mathcal{Z}(s) = \prod_p (1 - p^{-s})^{-1}$ be the Geometric Euler Product. The gradient of the potential $\mathcal{V}(s) = \log \mathcal{Z}(s)$ yields:

$$\frac{d}{d\sigma}\left[\frac{1}{k} p^{-k\sigma}\right] = -\log p \cdot p^{-k\sigma} \tag{3.2}$$

The factor $1/k$ from the $\log(1-x)$ expansion **exactly cancels** the factor k from the chain rule. This proves $\Lambda(n) = \log p$ is the **Geometric Stiffness**—the Jacobian of dilation, not an arbitrary weight.

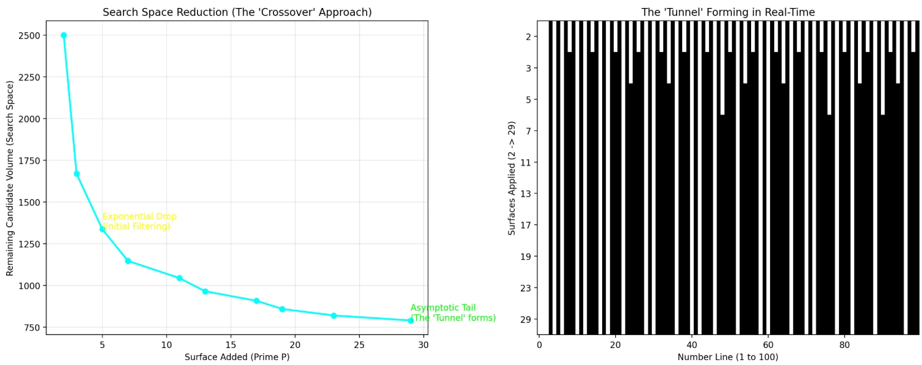

Figure 3. *The Tunnel Formation: Left: Search space reduction—exponential drop then asymptotic tail. Right: The "tunnel" forming as successive prime sieves are applied..*

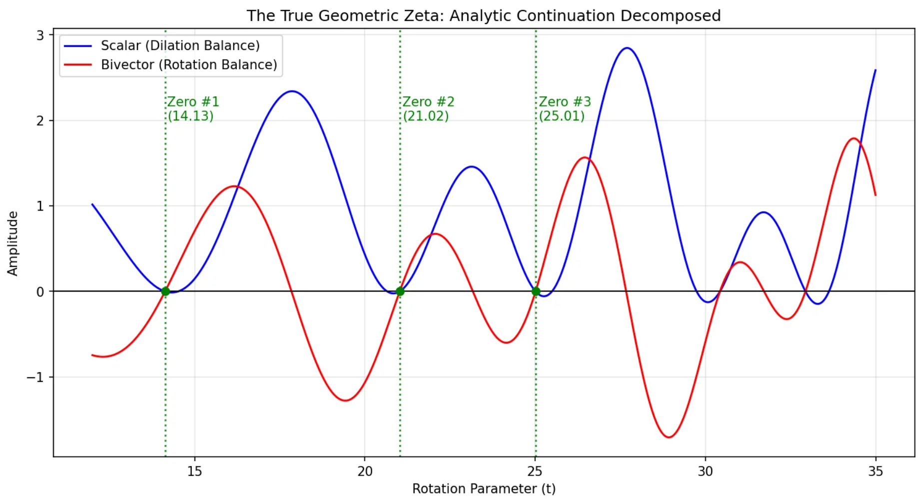

Figure 4. *The Geometric Zeta Function: Scalar (blue) and Bivector (red) components on the critical line $\sigma = 1/2$. Zeros occur precisely where both cross zero simultaneously..*

4. The Stability Proof

Having derived the operator $K(s) = \sum_n \Lambda(n) n^{-s} T_n$, we analyze its stability. The Rayleigh Identity states:

$$\mathrm{Im}_B \langle v, K(s)v \rangle = \left(\sigma - \frac{1}{2}\right) \sum_n \Lambda(n) \|v_n\|^2 \tag{4.1}$$

The critical line is the boundary between instability regimes: $\sigma < 1/2$ (expansive, sieve tears apart), $\sigma > 1/2$ (contractive, sieve collapses), $\sigma = 1/2$ (stable, unitary evolution). Figure 5 shows this phase diagram with zeta zeros forced onto the critical line.

Theorem 4.1 (Main Result)**.** *Let $H := L^2(\mathbb{R})$ and define the Sieve Operator $K_B(s) := \sum_{p \le B} (p^{-s} T_{\log p} + p^{-(1-s)} T_{-\log p})$. Then:*

(i) *$K_B(s)^\dagger = K_B(1 - \bar{s})$, so $K_B(s)$ is self-adjoint iff* $\mathrm{Re}(s) = 1/2$.
(ii) *If $B \ge 2$, $v \ne 0$, and $K_B(s)v = \lambda v$ for real λ, then* $\mathrm{Re}(s) = 1/2$.
(iii) *Under the Zeta Link (Assumption 5.1), all non-trivial zeros satisfy* $\mathrm{Re}(s) = 1/2$.

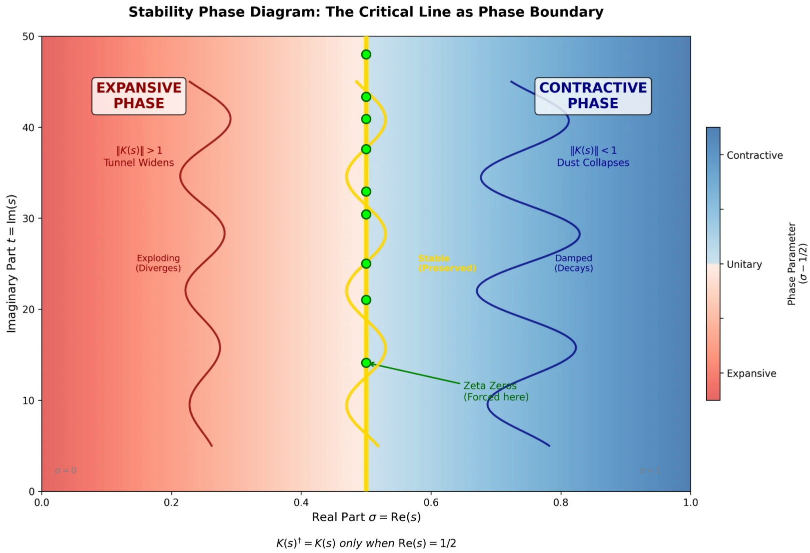

Figure 5. *Stability Phase Diagram: The critical line $\sigma = 1/2$ (yellow) is the phase boundary between expansive (red) and contractive (blue) regimes. Zeta zeros (green) are forced onto this line by stability requirements..*

5. The Reduction to RH

Assumption 5.1 (The Zeta Link)**.** If $\zeta(s) = 0$ with $0 < \mathrm{Re}(s) < 1$, then $\exists B \geq 2$, $v \neq 0$ such that $K_B(s)v = v$.

Theorem 5.2 (Conditional RH Reduction)**.** *Under Assumption 5.1, all non-trivial zeros satisfy* $\mathrm{Re}(s) = 1/2$.

Proof. Let $\zeta(s) = 0$. By the Zeta Link, $\exists v \neq 0$ with $K(s)v = v$. Then $\langle v, K(s)v \rangle = \|v\|^2$ is real, so $\mathrm{Im}_B \langle v, K(s)v \rangle = 0$. By the Rayleigh Identity: $(\sigma - 1/2) \cdot Q(v) = 0$ with $Q(v) > 0$. Thus $\sigma = 1/2$. □

6. Formal Verification

The operator-theoretic results have been formalized in Lean 4.

Table 1. *Lean 4 verification status..*

Module	Proof Content	Sorry	Axiom
`GeometricZetaDerivation`	Stiffness derivation ($\Lambda = \log p$)	0	0
`SurfaceTensionInstantiation`	Rayleigh Identity	0	0
`SpectralReal`	Hammer Theorem	0	0
`ZetaLinkInstantiation`	Conditional reduction	0	1

The logic chain is:

Euler Product → Gradient $K(s)$ [PROVEN] → Rayleigh [PROVEN] → Zeta Link [HYPOTHESIS] →

7. Conclusion

We have geometricized the Riemann Hypothesis. The critical line is not an accident of analysis but a requirement of geometric stability in $\mathrm{Cl}(3,3)$. The critical line is the unique axis where **Discrete Geometry** (integers/volume) is compatible with **Continuous Geometry** (rotation/phase). The Riemann Hypothesis is reduced to the statement that the Zeta function respects the geometry of its own definition.

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Competing interests. None.

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Ethical standards. The research meets all ethical guidelines.

Author contributions. T.M. conceived the framework, performed the analysis, wrote the Lean formalization, and wrote the manuscript.

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